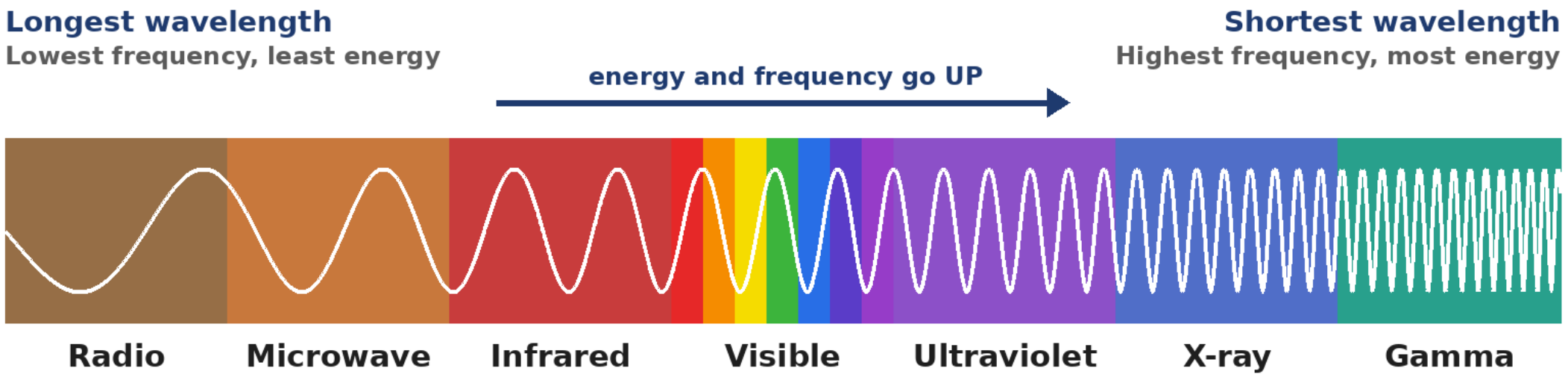
4.13 Unit 4 Review

Get ready for the 4.14 Unit 4 Mission



Today's Goals

I can explain that waves carry energy, not matter.

I can label a wave and use the 10 focus words.

I can compare transverse, longitudinal, mechanical, and EM waves.

I am ready for the Unit 4 Mission.

Rate each goal:

1 Not yet

2 Getting there

3 I can teach it

CASE #4 • THURSDAY

Did Toledo's water drop more than 10 feet?

WAVE WORD: *Seiche*

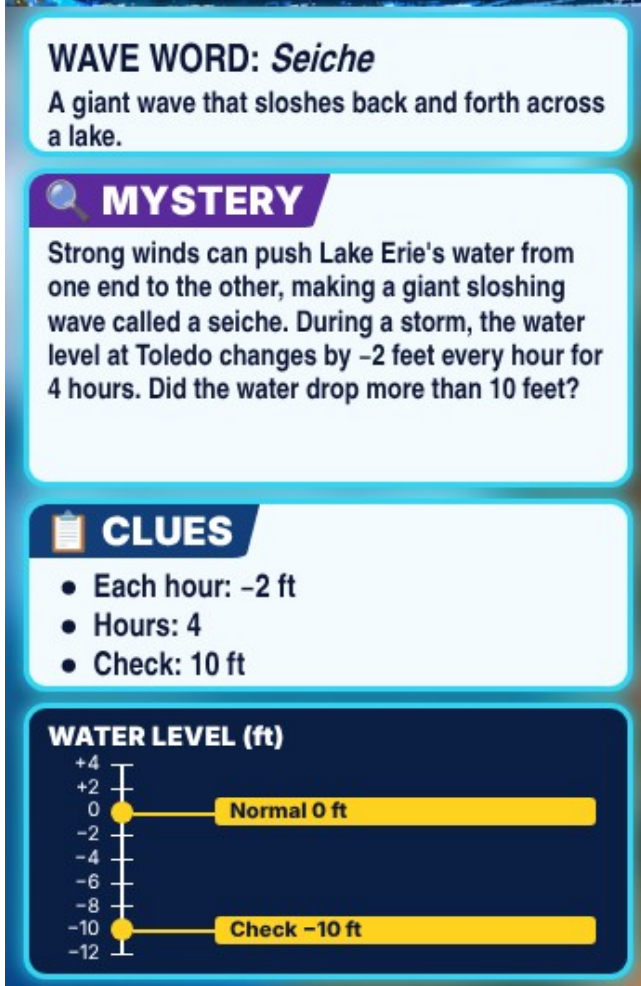
A giant wave that sloshes back and forth across a lake.

MYSTERY

Strong winds can push Lake Erie's water from one end to the other, making a giant sloshing wave called a seiche. During a storm, the water level at Toledo changes by −2 feet every hour for 4 hours. Did the water drop more than 10 feet?

CLUES

- Each hour: −2 ft
- Hours: 4
- Check: 10 ft





CLAIM

Did the water drop more than 10 feet?

☐ YES ☐ NO

EVIDENCE

Solve:

$(-2) \times 4 =$ _____

The water level changed by _____ ft.

REASONING

Finish the sentence:

The signs are _____, so the answer is _____. The water dropped _____ ft, which is _____ than 10 ft.

Today's Review Plan

4.13 Review

Classkick Q1–Q5 + Wordwall game

Vocab Review

Part 1 Picture Match + Part 2 Use the Words

Post Vocab survey

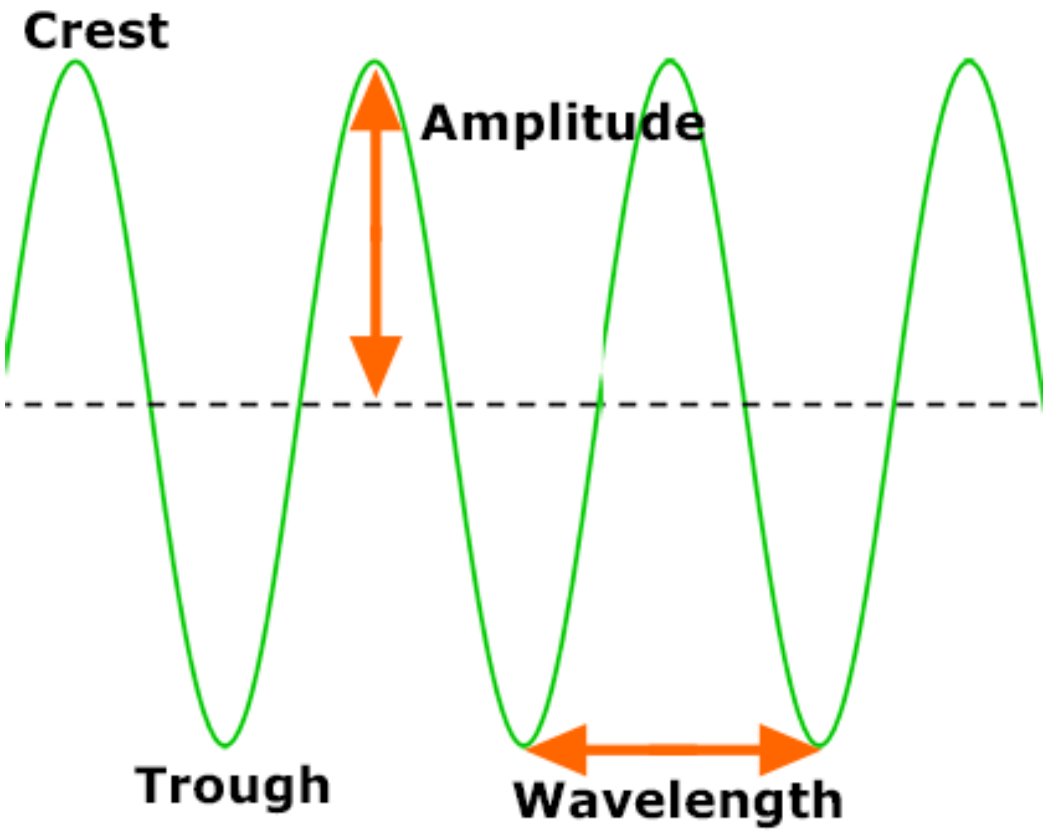
Rate the 10 words again. Closes Thu, Dec 10.

Mission Map

We fill it in together. Use it on the Mission.

The 4.14 Unit 4 Mission opens Mon, Dec 7. Regular close: Mon, Dec 14. Extended: Mon, Dec 21.

Parts of a Wave



Word	Meaning
Crest	the highest point
Trough	the lowest point
Wavelength	crest to crest (or trough to trough)
Amplitude	rest line to crest. Bigger = more energy
Frequency	waves per second

Don't fall for it: crest to trough is NOT a wavelength.

The 10 Focus Words

wave carries energy, not matter	**medium** the matter a wave travels through	**energy** the ability to do work or cause change	**frequency** waves that pass a point each second	**wavelength** crest to crest
amplitude how far the medium moves from rest	**transverse wave** medium moves at a right angle to the wave	**electromagnetic wave** can travel through empty space, like light	**refraction** a wave bends in a new medium	**density** how much mass is packed into a space

Compare: Kinds of Waves

	Transverse (up and down)	Longitudinal (back and forth)
Mechanical (needs a medium)	guitar string, jump rope	sound, slinky pushed in and out
Electromagnetic (crosses empty space)	light, radio waves, X-rays	never! EM waves are always transverse

Every wave gets TWO labels: how it moves (transverse or longitudinal) AND what it needs (mechanical or electromagnetic).

Review Poll

Which wave is BOTH longitudinal AND mechanical?

Single choice poll

A. Sunlight

Single choice poll

B. A shaken jump rope

Single choice poll

C. The sound of a drum

Single choice poll

D. Radio waves

Compare: The Density Flip

Light (EM wave)

SLOWER in denser stuff

fastest in air → slower in water → slower in glass

When it slows down at an angle, it bends: refraction (a straw looks broken).

Sound (mechanical wave)

FASTER in solids

slowest in air → faster in water → fastest in steel

Packed particles bump the next ones quickly.

Refraction: a wave bends when its speed changes in a new medium.

Review Poll

Which statement is TRUE?

Single choice poll

A. Light travels faster in glass than in air

Single choice poll

B. Sound travels faster in steel than in air

Single choice poll

C. Sound travels fastest in air

Single choice poll

D. Light needs a medium to travel

The Big Rules

Energy, not matter

The medium moves in place and returns. Only energy travels.

Amplitude

Bigger amplitude = more energy. It does NOT change the speed.

Frequency ↔ wavelength

One goes up, the other goes down.

Speed

Only the medium changes wave speed.

EM spectrum: radio → microwave → infrared → visible → UV → X-ray → gamma (more energy →)

Review Poll

You shake a rope faster, so the frequency goes UP. What happens to the wavelength?

Single choice poll

A. It gets longer

Single choice poll

B. It gets shorter

Single choice poll

C. It stays the same

Single choice poll

D. The wave gets taller

Who Is Right?

Why are gamma rays more dangerous than radio waves?

Ava says:

"Gamma rays are the biggest waves, so they hit harder."

Ben says:

"Gamma rays travel faster than radio waves."

Cam says:

"Gamma rays have the highest frequency and shortest wavelength, so they carry the most energy."

Dee says:

"Gamma rays need a medium, and radio waves do not."



A. Ava



B. Ben



C. Cam



D. Dee

Play: Review Waves and EMS

Wordwall review game

1. Play the game.
2. Then answer on Classkick:

Why can we see the Sun but not hear it?

▶️ Open the Wordwall review

Light is an electromagnetic wave. It can travel through empty space.

Sound is a mechanical wave. It needs a medium, and space has almost none.

Vocab Review: Parts 1 and 2

Part 1: Picture Match

Use the picture and the meaning. Drag the focus word into each box.

▶️ Open Wave Vocab Flashcards (SERP)

Part 2: Use the Words

Odd one out, the concert story, and explain one word to a 5th grader.

▶️ Open Wave Diagram Labeling

Explain it to a 5th grader: Amplitude is how tall a wave is from the middle line. A big ocean wave has a big amplitude and more energy, so it can knock you over.

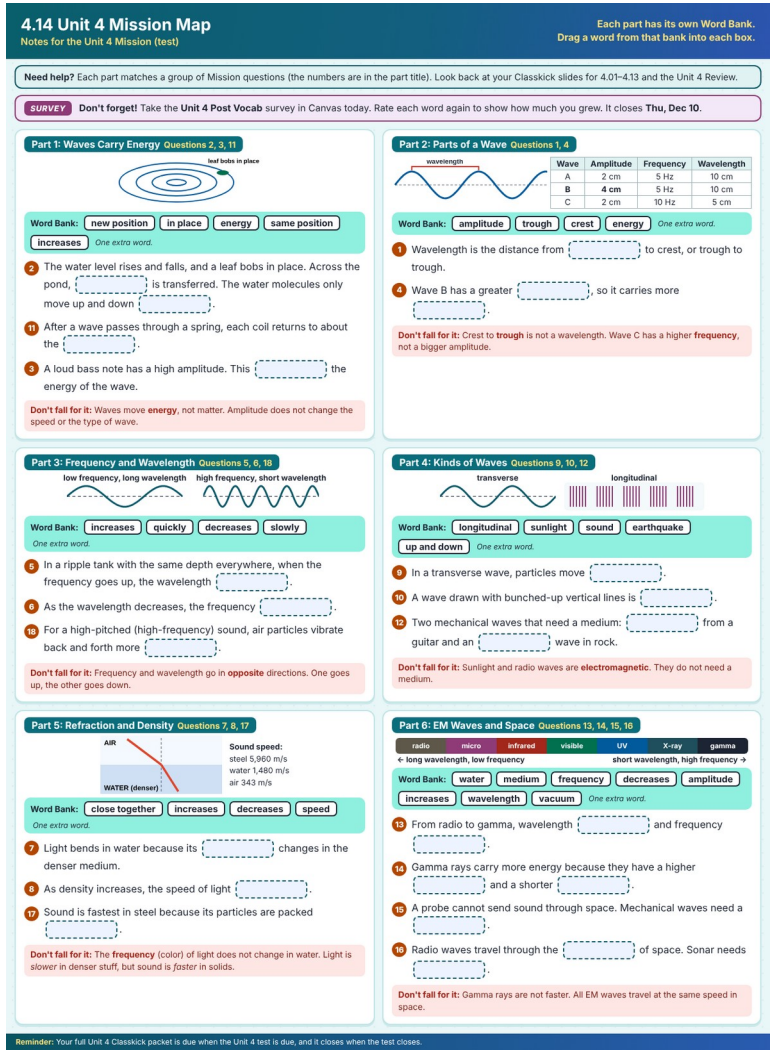
Before the Mission

Take the Unit 4 Post Vocab survey

Rate each of the 10 words again to show how much you grew. It closes Thu, Dec 10.

wave	**medium**	**energy**	**frequency**	**wavelength**
amplitude	**transverse wave**	**electromagnetic wave**	**refraction**	**density**

4.14 Mission Map: Together



4.14 Unit 4 Mission Map

Notes for the Unit 4 Mission (test)

Each part has its own Word Bank. Drag a word from that bank into each box.

Need help? Each part matches a group of Mission questions (the numbers are in the part title). Look back at your Classkick slides for 4.01–4.13 and the Unit 4 Review.

SURVEY **Don't forget!** Take the **Unit 4 Post Vocab** survey in Canvas today. Rate each word again to show how much you grew. It closes **Thu, Dec 10**.

Part 1: Waves Carry Energy Questions 2, 3, 11

leaf bobs in place

Word Bank: new position | in place | energy | same position | increases — *One extra word.*

2. The water level rises and falls, and a leaf bobs in place. Across the pond, _____ is transferred. The water molecules only move up and down _____.
11. After a wave passes through a spring, each coil returns to about the _____.
3. A loud bass note has a high amplitude. This _____ the energy of the wave.

Don't fall for it: Waves move **energy**, not matter. Amplitude does not change the speed or the type of wave.

Part 2: Parts of a Wave Questions 1, 4

wavelength

Wave	Amplitude	Frequency	Wavelength
A	2 cm	5 Hz	10 cm
B	4 cm	5 Hz	10 cm
C	2 cm	10 Hz	5 cm

Word Bank: amplitude | trough | crest | energy — *One extra word.*

1. Wavelength is the distance from _____ to crest, or trough to trough.
4. Wave B has a greater _____, so it carries more _____.

Don't fall for it: Crest to **trough** is not a wavelength. Wave C has a higher **frequency**, not a bigger amplitude.

Part 3: Frequency and Wavelength Questions 5, 6, 18

low frequency, long wavelength | high frequency, short wavelength

Word Bank: increases | quickly | decreases | slowly — *One extra word.*

5. In a ripple tank with the same depth everywhere, when the frequency goes up, the wavelength _____.
6. As the wavelength decreases, the frequency _____.
18. For a high-pitched (high-frequency) sound, air particles vibrate back and forth more _____.

Don't fall for it: Frequency and wavelength go in **opposite** directions. One goes up, the other goes down.

Part 4: Kinds of Waves Questions 9, 10, 12

transverse | longitudinal

Word Bank: longitudinal | sunlight | sound | earthquake | up and down — *One extra word.*

9. In a transverse wave, particles move _____.
10. A wave drawn with bunched-up vertical lines is _____.
12. Two mechanical waves that need a medium: _____ from a guitar and an _____ wave in rock.

Don't fall for it: Sunlight and radio waves are **electromagnetic**. They do not need a medium.

Part 5: Refraction and Density Questions 7, 8, 17

AIR | WATER (denser)

Sound speed: steel 5,960 m/s; water 1,480 m/s; air 343 m/s

Word Bank: close together | increases | decreases | speed — *One extra word.*

7. Light bends in water because its _____ changes in the denser medium.
8. As density increases, the speed of light _____.
17. Sound is fastest in steel because its particles are packed _____.

Don't fall for it: The **frequency** (color) of light does not change in water. Light is *slower* in denser stuff, but sound is *faster* in solids.

Part 6: EM Waves and Space Questions 13, 14, 15, 16

radio | micro | infrared | visible | UV | X-ray | gamma

← long wavelength, low frequency — short wavelength, high frequency →

Word Bank: water | medium | frequency | decreases | amplitude | increases | wavelength | vacuum — *One extra word.*

13. From radio to gamma, wavelength _____ and frequency _____.
14. Gamma rays carry more energy because they have a higher _____ and a shorter _____.
15. A probe cannot send sound through space. Mechanical waves need a _____.
16. Radio waves travel through the _____ of space. Sonar needs _____.

Don't fall for it: Gamma rays are not faster. All EM waves travel at the same speed in space.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

6 parts, one for each group of Mission questions

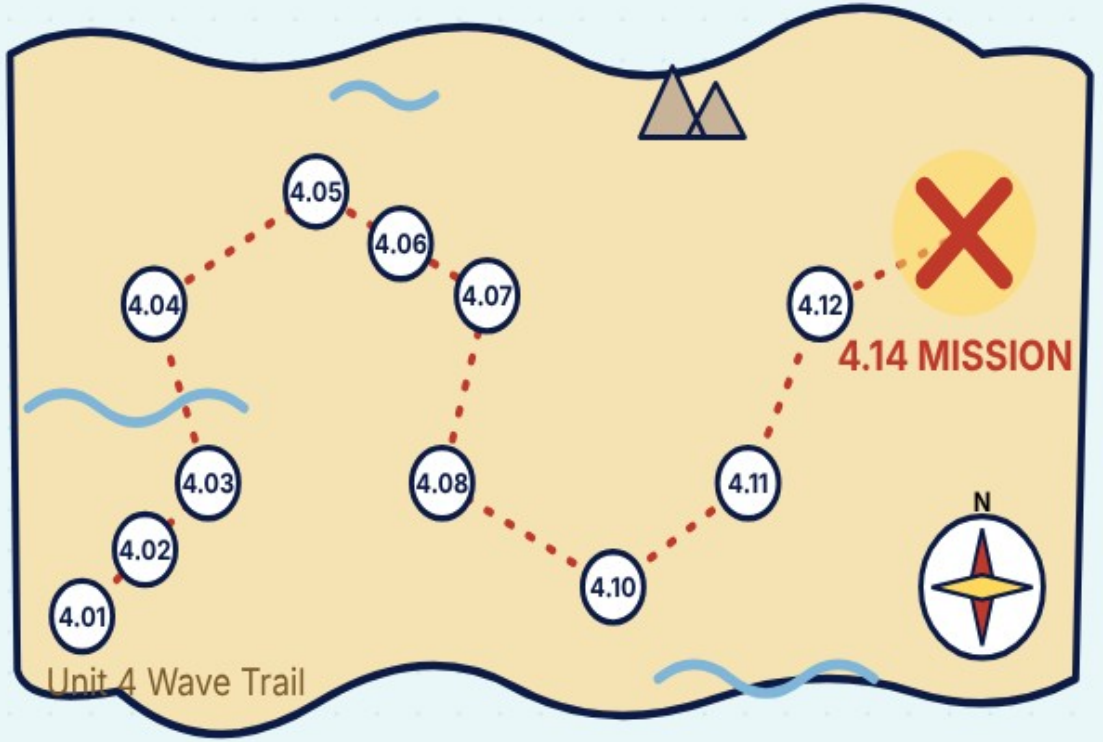
1. Waves carry energy (Q2, 3, 11)
2. Parts of a wave (Q1, 4)
3. Frequency and wavelength (Q5, 6, 18)
4. Kinds of waves (Q9, 10, 12)
5. Refraction and density (Q7, 8, 17)
6. EM waves and space (Q13–16)

Your Final Mission Awaits!

4.14 Unit 4 Mission · Waves Test

Every good adventurer takes a **map** on an important mission!





MISSION DUE
Mon, Dec 14

EXTENDED TIME
Mon, Dec 21

Don't wait until the trail goes cold! Finish by the due date.

The test is open book and open notes!

Pack your gear: your 4.14 Mission Map, Classkick notes, and Unit 4 vocabulary pages.

Key Ideas

1. Waves carry energy, not matter.
2. Transverse = up and down. Longitudinal = back and forth. Mechanical needs a medium; EM does not.
3. Frequency up = wavelength down. Bigger amplitude = more energy.
4. Only the medium changes speed. Light: slower in denser stuff. Sound: fastest in solids.

Re-rate your goals. Mission opens Mon, Dec 7. You've got this!